Observational Incompleteness: From Deterministic Dynamics to Quantum Observations

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Abstract

Could the randomness of quantum measurements arise because observers cannot access all the information in an underlying deterministic reality? Observational incompleteness (OI) is the hypothesis that every physical observer is a proper part of a finite reversible deterministic system and records only the visible marginals of its dynamics. At every finite horizon, three classes of law coincide: all finite stochastic laws, the visible marginals of finite reversible dynamics, and the records of finite unitary systems measured in one fixed basis; the known components (Egri et al.; Barandes) are stated as one machine-checked theorem. Because Markov laws lie in every class, the discriminating content is memory: every faithful realization carries the law's conditional memory in its hidden state. Stochastic, quantum-like records can therefore arise exactly from deterministic dynamics, and any memory they show must be stored in the hidden part. In a finite reversible world whose hidden odds are shared by every visible start and drawn independently of it, the forecasts from every start, starts of probability zero included, are P-indivisible somewhere in the recurrence cycle whenever some table of forecasts from the start is not a permutation, in particular whenever a visible step is not fully predictable from the current observation; with every start of positive probability, this is equivalent to conditional memory at some order below the recurrence time. For an operational theory with a stated minimal structure on a nonempty finite system, the available operations, on the system and with every finite ancilla attached, are exactly the normalized finite Kraus instruments if and only if five completion conditions hold, and OI alone does not imply them. Applied to physics, OI places the zero-point sum and the effective cosmological constant at different levels of description, and separates conservation of black-hole information from its accessible recovery. Open are the completion conditions a physical substratum supplies, the Born exponent and the gravitational field equations.

1. Introduction

1.1 Embedded observation

Could the randomness of quantum measurements arise because observers cannot access all the information in an underlying deterministic reality? Einstein's objection to quantum randomness is usually paraphrased as "God does not play dice". The picture examined here replaces dice with

cards: God plays cards. A deck evolves by fixed rules, nothing in the rules is random, and yet a player who sees only some of the cards finds the game uncertain. The picture is an analogy. It shows how hidden information, deterministic rules and random-looking observations fit together; it proves nothing about quantum mechanics, and every result below is stated and proved without it.

In the simplest version two cards lie on a table, one face up and one face down, and each shuffle swaps them. After one shuffle the player cannot predict the colour now showing any better than by tossing a coin. After a second shuffle the original card is back, with certainty. The player's record therefore has memory: what is seen next depends on what was seen earlier, not only on what is showing now, and that memory is kept in the face-down card. After the unseen deal nothing random happens; the uncertainty is the player's, and it comes from what the player cannot see. The exact table is in §2.6, and the game's statistics are those of ordinary classical probability.

So framed, a question about interpretation becomes three mathematical questions. Which patterns of observations can a partly hidden deterministic world produce? Where must the hidden information sit? What does it force the observations to look like? Sections 2 and 3 answer them for finite worlds and state what the answers leave open.

A *bijection* sends each configuration to exactly one successor and never sends two configurations to the same one, so it loses no information: the whole present determines the whole past. What the observer sees of the dynamics, the hidden part being ignored, is its *visible marginal*. An observer inside a closed system registers only part of its configuration. If the whole evolves by a bijection, hidden states compatible with the same visible state can send it to different futures, so the visible sector evolves stochastically although the whole evolves deterministically.

This is modelled by a triple (S, φ, V) [Main §1.2]: a finite total system $S = \mathcal{C}_V \times \mathcal{C}_H$ of visible configurations $\mathcal{C}_V$ and hidden configurations $\mathcal{C}_H$, a bijection φ of S, and an observer V that registers $x \in \mathcal{C}_V$ and not $h \in \mathcal{C}_H$. The bijection is the reversible representative of the hidden dynamics, motivated by observed near-unitarity, and finiteness of all of S is posited [Main §1.2, Remark (status of the early commitments)]. The finite reversible system (S, φ) is the *substratum*.

Observational incompleteness (OI) is the physical hypothesis that every physical observer is in this situation: a part of a finite, reversible, deterministic world with a specified readout and preparation. The hidden coordinates are not directly read at that instant. Whether they can be inferred depends on the retained history and permitted operations, not on embedding alone. In the known swap game, the first two visible readings already identify both cards, even though the visible process has memory. More generally, a quantity is identified exactly when it is constant on the initial states compatible with the retained record [Substratum, Prop. 24.5]. It is not joint unsharpness of non-commuting observables: the restriction can arise for a single fixed readout, and no uncertainty relation constrains fixed-readout visible laws [Main §3.4].

What does the substratum run on? Hawking's story of a world resting on a turtle, and then on turtles all the way down, asks a different question from the cards: what supports the hidden construction itself? An exactly decoupled deeper register can be added without changing the specified visible law, under the preparation assumptions of [Substratum, Theorem 24(iii)]. For that observational interface it supplies another description of the same data. A deeper sector that can couple back may instead change later records and cannot be dismissed on that ground. The restricted invariances do not classify every possible deeper implementation or prove that the regress ends. A finite bijection is the model being investigated, not a demonstrated ultimate ontology [Substratum §§3–4; Book §3.8].

1.2 The idea and its consequences

Hiding part of a reversible world can make what is seen random, give it memory, and make its probabilities impossible to chain step by step. The stochastic–quantum correspondence relates these features to quantum theory [Bar23; Bar25], and over any finite number of steps it is exact and universal (Theorem 1, §2.2): every finite pattern of random-looking records, every pattern of quantum records from one fixed measurement among them, is what some finite reversible deterministic world shows to an observer who sees only part of it. Apparently stochastic, quantum-like observations can therefore arise exactly from deterministic dynamics. Because this holds for every pattern, the discriminating content lies in where the memory of a pattern must be kept (§2.4) and in what the world's eventual return to its starting configuration does with it (§2.5). Section 3 states how far the correspondence reaches toward operational quantum mechanics, §§4–5 apply OI to vacuum energy and to black-hole information, and §6 sorts the results by status. The core finite-law and memory arguments are in [Main]; proofs and qualifications also occur in the companion papers, book appendices and verification records, with the relevant source identified where used. Gravitational applications are in [GR] and [Book].

Contributions. The paper is expository and claims no new theorem; each known component is attributed where it is stated (§§2.2, 2.4, 2.5). The results of §§3.2–3.5 are proved in [GR] and [Main], the Bell ceiling being Bell's theorem under the framework's hypotheses, and those of §5 in unpublished research records with exact or exhaustive checks [BHa; BHb]; the novelty of the §§3.2–3.3 and §5 results has not been assessed. Deterministic systems have been analysed with quantum methods before, for example in [tH14].

Conventions. "Machine-checked" means checked in Lean 4/Mathlib [dMU21; Mathlib20] at commit `17b830bc` by the recipe under "Code, data and formal verification" (declarations in Appendix A); "proved on paper" means a written proof without a machine check; an "exact probe" is an exhaustive computation in exact rational arithmetic at small size. Where a subsection's evidence classes and attributions are long, they close it in a paragraph labelled *Proofs and sources*; elsewhere they are given inline. A *faithful realization* of a law is a bijection φ of $\mathcal{C}_V \times \mathcal{C}_H$ with an initial law under which the visible trajectory has that law.

2. Theoretical foundation: the established results and their assumptions

2.1 Setting

Fix finite sets $\mathcal{C}_V$ (with $|\mathcal{C}_V| \geq 2$) and $\mathcal{C}_H$, a bijection φ of $\mathcal{C}_V \times \mathcal{C}_H$, and a horizon K. A law is a probability distribution on visible trajectories $(x_0, \dots, x_K) \in \mathcal{C}_V^{K+1}$. In a realization, X_t and H_t are the visible and hidden coordinates at time t, $X_{<t} = (X_0, \dots, X_{t-1})$ is the visible past and $X_{>t} = (X_{t+1}, \dots, X_K)$ the visible future. Information is measured in bits, $\|\mu - \nu\|_{\mathrm{TV}} = \frac{1}{2}\sum_k |\mu_k - \nu_k|$ is the total-variation distance, and $g_{\#}\mu$ is the pushforward of a law μ under a map g.

In words: a *law* lists the probability of every possible sequence of observations over $K+1$ steps, and a *realization* of a law is a hidden deterministic world that reproduces it. A law is *Markov*, or memoryless, when the next observation depends only on the present one. A bijection of a finite set eventually repeats: its *order* L, the world's *recurrence time*, is the least number of steps after which every configuration has returned to itself, and one such return is a *recurrence*. A *stochastic matrix* is a table of transition probabilities with one row per starting state, each row nonnegative and summing to one; a *permutation matrix* is the stochastic matrix in which every row is a single certainty and no two rows agree. The *total-variation distance* between two probability lists is the

largest difference in probability they assign to any one event: 0 for identical lists, 1 for lists that give no outcome positive probability in common. The *conditional mutual information* $I(A; B \mid C)$ measures, in bits, how much A still tells about B once C is known; it is zero exactly when, given C, A carries no further information about B.

The visible law of a realization depends on the dynamics, the partition and the initial law (the prior), which is part of the realization datum, not selected by invariance under φ; the counting measure is a maximal-entropy default [Main §1.2, Remark (the measure as a realization datum)].

Three classes of law are compared [Main §3.4]:

- S_K: all probability distributions on $\mathcal{C}_V^{K+1}$.
- D_K: the visible trajectory laws of a bijection of $\mathcal{C}_V \times \mathcal{C}_H$, for some finite $\mathcal{C}_H$, under an initial law that may correlate the visible and hidden coordinates.
- Q_K: the *fixed-basis Born records*, the laws of finite-dimensional unitary systems read out by projective Born measurement in one fixed basis at every step, from an initial state diagonal in that basis, the record being a fixed function of each outcome.

Informally, S_K is every statistical pattern whatever; D_K is every pattern that a partly hidden, deterministic, reversible world can show; and Q_K is every pattern of results from a finite quantum system that evolves by a *unitary*, the reversible, probability-preserving evolution of quantum theory, and is measured in the same way at every step, each result having the probability the *Born rule* assigns, the squared magnitude of its *amplitude*, the complex number quantum theory attaches to that result.

2.2 The finite-horizon correspondence

Which patterns of observations can a partly hidden deterministic world produce, and how do they compare with what quantum measurements produce? Over any fixed number of steps the answer is that the three lists coincide.

Theorem 1 (finite-horizon stochastic–reversible–unitary equivalence [Main §3.4]). *For every finite visible alphabet and every finite horizon K,*

$$\mathsf{S}_K = \mathsf{D}_K = \mathsf{Q}_K,$$

so each class is the set of all probability laws on $\mathcal{C}_V^{K+1}$.

That is, any pattern at all can be produced by a hidden deterministic world, and also by a quantum system measured the same way at every step. Finite quantum records of this fixed-measurement kind therefore carry no statistical signature that a partly hidden deterministic world could not reproduce, and conversely. This settles a question of possibility, not of fact: it does not show that nature is such a world.

One construction is simple to picture. The hidden register holds a whole sequence drawn in advance from the law, together with a clock, and each step brings the next entry into view; the dynamics stays a bijection, and the observer, who sees only the entries already shown, finds them random with exactly the prescribed probabilities.

Realization. For $\mathsf{S}_K \subseteq \mathsf{D}_K$ the realization stores a sampled trajectory, with a clock, in the hidden register [Main §3.4, Theorem (finite-horizon stochastic–reversible–unitary equivalence); EGGH26, Prop. 13]. Every law also has a realization whose hidden prior is independent of x_0, by the finite-horizon process dilation for rational conditional kernels at positive-probability histories, the visible

initial law being arbitrary, and by its response-table extension for arbitrary kernels (proved on paper, with exact probes) [Main §3.4, Theorem (finite-horizon process dilation) and its Extension].

Quantum representation. For $\mathsf{D}_K \subseteq \mathsf{Q}_K$ the representation is the permutation unitary $U_\varphi|x,h\rangle = |\varphi(x,h)\rangle$, read out in the configuration basis at every step [Main §3.4]; $\mathsf{Q}_K \subseteq \mathsf{S}_K$ is the Born rule.

Proofs and sources. Theorem 1 is machine-checked: the three inclusions $\mathsf{S}_K \subseteq \mathsf{D}_K \subseteq \mathsf{Q}_K \subseteq \mathsf{S}_K$ and their compositions, with no construction entering as a hypothesis (Appendix A).

Attribution. The trajectory-labelling realization is [EGGH26, Prop. 13], packaged as one iterated bijection; the independent-prior form is their Prop. 16, without reversibility, and its Markov one-step case is the random-mapping representation [LP17, Prop. 1.5]. A permutation dynamics is a special case of a unistochastic process, one with transition probabilities $|U_{ij}|^2$ for a unitary U [Bar25, p. 12 of arXiv v3; Bar23 §4.3; Bar26 §4]; the one-step $|\mathcal{C}_V|^2$-dimensional ancilla dilation of [Main §3.1] is that of [Bar26, Thm 3.3; Sch21, Thm 2]; and the every-step record law is [SGD19, Eq. (3)]; whether such records remain consistent when measurements are omitted [SGD19; Milz20] does not arise for Q_K, which measures at every step.

2.3 Scope of the correspondence

What does a correspondence that holds for every pattern leave out?

Markov laws lie in all three classes, so neither a finite reversible realization nor a fixed-basis unitary representation distinguishes a law with memory from one without [Main §3.4, Remark (what this equivalence does and does not say)].

The relation $p = |U|^2$ is part of the definition of Q_K, not an output. The unitaries supplied are permutations, with entries 0 and 1, so the recovered table is the same for $|U|^q$ at every $q > 0$, and the quadratic exponent is not selected [Main §3.4; Bar26, Thm 3.1].

A finite bijection has finite order L, so one fixed realization has an L-periodic multi-time law: the correspondence holds per horizon, and a nonperiodic law has no exact all-time realization by one fixed finite substratum [Main §3.4].

The correspondence thus answers the question of possibility: a hidden deterministic model exists for every law, so its existence says nothing about a particular law, and one such world reproduces a pattern over a chosen finite stretch of time, not, in general, forever. Because the match is exact for every pattern, no record of one repeated measurement can by itself rule out a partly hidden deterministic account, and the memory results that follow apply to every such quantum record. The content lies in what such worlds must contain when the record has memory (§§2.4–2.5).

2.4 Where the memory lives

When the observed record has memory, so that what comes next depends on more than what is visible now, where is that memory kept? In the card game the record remembers the dealt card after the first shuffle, although the face-up card no longer shows it, because the face-down card holds it. The same is true in every hidden deterministic world that reproduces a record: whatever the record remembers is held in the hidden part, and the hidden part must be large enough to hold it.

Write $M_t := I(X_{<t}; X_{t+1} \mid X_t)$ for the conditional memory at time t: the information about the

next observation that the observed past carries beyond what the observed present already tells. A law is *Markov*, that is memoryless, exactly when $M_t = 0$ at every t. The *hidden predictive memory* $I(X_{<t}; H_t \mid X_t)$ is the information the hidden state holds about the observed past beyond what the observed present tells; it is at least the conditional memory (Theorem 2(c)), and only part of it need affect what is seen next. In a realization, write $f_x(h) = \pi_V(\varphi(x, h))$, with π_V the projection onto $\mathcal{C}_V$, and μ_ω for the hidden posterior, the probabilities of the hidden states given what has been seen, after a positive-probability visible history $\omega = (x_0, \dots, x_t)$ ending at $x_t = x$.

Theorem 2 (unavoidable hidden predictive memory [Main §3.4]). *In every faithful realization:*

(a) $P(X_{t+1} \mid \omega) = (f_x)_{\#}\mu_\omega$*;*

(b) for histories ω, ω' *ending at* x*,*

$$\|P(X_{t+1} \mid \omega) - P(X_{t+1} \mid \omega')\|_{\mathrm{TV}} \le \|\mu_\omega - \mu_{\omega'}\|_{\mathrm{TV}};$$

(c) $M_t \le I(X_{<t}; H_t \mid X_t) \le \log_2 |\mathcal{C}_H|$.

That is: the next observation depends on the past only through what the past reveals about the hidden state (a); two pasts can lead to different predictions only if they leave at least as different odds on the hidden state (b); and the memory visible in the record never exceeds the hidden memory, which never exceeds the hidden sector's capacity in bits (c).

Predictive memory and state inference. Theorem 2 bounds information useful for predicting the next output; it does not say that the hidden state stays uncertain after the full history is retained. In the swap game the past reading tells the player the current face-down card exactly. For a known finite permutation and a specified preparation support, Proposition 24.5 of [Substratum] gives the general record-fibre criterion: collect the initial states producing the retained record; a quantity is known exactly when it has the same value on that set. One candidate identifies the full state; for the visible/hidden product readout, the current hidden state has exactly the same singleton criterion. Passive future-trace classes stabilize after at most $\max(1, N-1)$ readings on an N-state carrier, but those classes need not be singletons. The constant-on-fibre criterion, the initial/current-state and current-hidden singleton criteria, and the two-reading swap identification are machine-checked for nonempty compatible-state sets. The Bayesian support step and the $N-1$ stabilization bound remain proved on paper, with exact finite controls. None of these results supplies a physical storage, preparation or intervention mechanism.

A *memory witness* at t is a pair of positive-probability histories ending at the same x_t with different next-step laws; by the vanishing criterion for conditional mutual information [CT06], a law has one exactly when $M_t > 0$.

Corollary (proved on paper [Main §1.5]). $M_t > 0$ holds exactly when every faithful realization carries readout-relevant hidden memory at t, that is, a pair of posteriors with different pushforwards. The forward direction rests on Theorem 2(a) in each realization; the converse on Theorem 2(a) in one realization, which Theorem 1 supplies (`S_imp_D`); both use the standard vanishing criterion for conditional mutual information [CT06], which, like the time-t identification, is not machine-checked.

So the record has memory exactly when every hidden deterministic world that reproduces it keeps, in its hidden part, a trace of the past that changes what is seen next: readout-relevant hidden memory, the part of the hidden predictive memory that supports the observed process's dependence on history. Conversely, if any one hidden deterministic world reproducing the record keeps such a trace, the record has memory. Keeping the past is not the same as memory: every reversible world

run from the start stores its whole visible past, even for a memoryless record, and memory is the part of that store that changes what is seen next.

The hidden sector also has a minimum size and a minimum memory. With $I^* = \max_t I(X_{<t}; X_{>t} \mid X_t)$, a realization of the law needs $|\mathcal{C}_H| \geq 2^{I^*}$ (machine-checked in the joint-law form of Theorem 2). Over deterministic completions, not required to be bijective, the least hidden predictive memory $I(X_{<t}; H_t \mid X_t)$ is exactly $I(X_{<t}; X_{>t} \mid X_t)$, attained by the canonical predictive quotient, and equals M_t exactly when $I(X_{<t}; X_{>t+1} \mid X_t, X_{t+1}) = 0$, while bijective realizations run from time 0 carry $H(X_{<t} \mid X_t)$, Markov laws included [Main §3.4, Remark (a variational question)]; the lower bound is machine-checked for bijective realizations and proved on paper for non-injective ones. Conversely, memory has a floor: if two values of $X_{<t}$ followed by a common present state X_t, each of joint probability at least p_0, induce next-step laws at total-variation distance at least δ, then $M_t \geq p_0\delta^2/\ln 2$, by Pinsker's inequality (proved on paper, with exact probes) [Main §2.3, Lemma (accessible conditional memory from readback); CT06]. Observed memory thus bounds a hidden-sector quantity: measurable conditional information fixes a lower bound on the hidden sector's size and the exact minimum of the hidden predictive memory a deterministic completion must carry.

For actual finite-permutation trajectories, the capacity bound is also machine-checked uniformly in time and in every finite future block. With $I^*_{\mathrm{fin}} = \sup_{t,L\in\mathbb{N}} I(X_{<t}; (X_{t+1}, \ldots, X_{t+L}) \mid X_t)$, the checked conclusion is $I^*_{\mathrm{fin}} \leq \log_2 |\mathcal{C}_H|$, hence $2^{I^*_{\mathrm{fin}}} \leq |\mathcal{C}_H|$. This is a genuine supremum over all finite blocks, for one fixed permutation and initial law. Identifying it with conditional information about a literal infinite future requires a further argument; the formal result makes no such identification. The time-t identification mentioned in the Corollary refers to its conditional-pushforward use of Theorem 2(a), which is separate from this checked information bound.

Four conditions on a hidden sector are named in [Main §1.3], as diagnostics of a realization rather than hypotheses of Theorem 1: (C1) influence; (C2) persistence of records; (C3) capacity, $\log_2 |\mathcal{C}_H| \geq I^*$; and (C4) history readback, a memory witness at some accessible order, an order short compared with the recurrence time, as accessible windows are [Main §2.3, Scope]. In a faithful realization (C4) coincides with $M_t > 0$, and the Corollary places it in the hidden state; (C1) follows from (C4), and (C3) holds by the capacity bound. (C2) has two forms. (C2-structural), the persistence of a stored record until its readback, holds by reversibility in every bijective realization and constrains coarse-grained descriptions of the hidden sector; (C2-slow), a hidden coarse-grained mixing time long compared with the accessible window, is one mechanism for it, and fast hidden dynamics that conserves the record is another [Main §1.3]. The necessity of (C2) is a theorem in the coupling-step model, in which the next visible state is a function of the current visible and hidden states and π_H is the model's reference hidden law, toward which a mixing hypothesis would have the hidden state relax between couplings [Main §3.4, Theorem (physical memory: C2, process form)]. There, observed conditional memory $M_k > 0$ at an accessible order k forces some realized hidden conditional at that coupling time to lie at total-variation distance at least $\varepsilon_{\min}(M_k, |\mathcal{C}_V|) > 0$ from π_H, with no mixing hypothesis; under the mixing hypothesis this becomes the slow-bath form, a lower bound on the mixing time.

So the four conditions are a checklist for a hidden sector that produces observable memory: it must influence what is seen (C1), keep what it records (C2), have room to record it (C3), and feed the record back into what is seen (C4). Coupling, persistence and capacity without readback produce no memory: a fair coin has realizations with all three and zero conditional memory on arbitrarily long horizons [Main §2.3, Remark (why (C4) is a separate condition)]. In the coupling-step model,

memory in the record at an accessible order means that the hidden part has not settled to its reference odds at that coupling time. The memory results establish that observed memory is real information held out of sight: it must be stored in the hidden part, it sets a minimum size for that part, and it is what separates laws with memory from memoryless ones, which the correspondence of §2.2 cannot.

Proofs and sources. *Theorem 2.* Part (b) and the first inequality of (c) are data-processing inequalities. (a)–(c) are machine-checked for a joint law of history, present and hidden state. In addition, (c) is machine-checked for the actual trajectory obtained by pushing an arbitrary initial law through one finite permutation. The conditional-pushforward and total-variation forms (a)–(b), and the vanishing criterion used in the Corollary, retain their written time-t arguments [Main §3.4, Theorem (unavoidable hidden predictive memory); CT06].

2.5 Recurrence and indivisibility

If the weather had no memory, a forecast from Monday to Wednesday could be built from one from Monday to Tuesday and one from Tuesday to Wednesday. When can the probabilities of an observed process be chained in this way? In the card game the forecast from the deal to the first shuffle is a coin toss whichever card was dealt, yet the forecast to the second shuffle is certainty; no forecast from the first shuffle to the second can turn two identical coin-toss rows into two different certainties, so the chain breaks. The property that fails is called *P-divisibility*: whether the forecasts from the start admit step-by-step intermediate forecasts over the interval in question. Suppose the dealer always deals red face up, so that a black start never appears in the player's sample. The question is still what the same shuffle, with the face-down card dealt the same way, at even odds, does from a red start and from a black start, and the black start belongs to that comparison although it never occurs. The comparison is about one fixed rule, the same hidden preparation tried from every visible start the world could have, not about the starts a sample happens to contain, and it means something only because the face-down card is dealt the same way whichever card is face up: if the dealer put the opposite colour face down, the forecast from red and the forecast from black would describe two different deals, not one rule started in two ways. A forecast from a start is a forecast of what the rule does only if the hidden part is set up by the same procedure every time; if the procedure changed with the start, a difference between two forecasts could come from the setup rather than from the dynamics.

Why keep a start that never occurs? Such a start is not impossible: the world can be started there; the dealer who always deals red simply never does. A different deck and shuffle shows why it still matters. Suppose its rule makes a red start always show red while a black start, with the same face-down odds, may show either colour, and the dealer always starts with red. What the observer actually sees is red at every step, with no uncertainty; what happens from a black start under that preparation is never observed. The rooted family, the table of forecasts from every start, still asks what the same dynamics and the same hidden preparation would produce from black: that forecast is part of the mathematical comparison, not something this observer measures. After the first shuffle both forecasts give red a positive chance, an overlap that comes only from the start that is never dealt [Main §2.3]. Both forecasts return to certainty at step 2, earlier than the whole world, which returns to its starting configuration only at its recurrence time, step 3. The failure of chaining comes from the step-1 overlap together with this step-2 return, not from the step-3 recurrence: no step-by-step forecast can carry the overlapping forecasts of step 1 to the two certainties of step 2, so the forecasts fail to chain over the first two steps, although no dealt game ever looks uncertain.

In a finite reversible world whose hidden part starts with the same odds whatever is visible at the start, any visible step that is not fully predictable from the current observation makes its failure, for the forecasts from every starting point, unavoidable somewhere in the world's cycle.

Fix one hidden prior shared by every root and drawn independently of it, and write Γ_t for the *rooted transition matrix* from time 0 to time t, with a row for every $x_0 \in \mathcal{C}_V$, roots of visible probability zero included. The family is *P-divisible on* $\{0, \dots, n\}$ if for all $s < t \leq n$ some stochastic $\Delta_{s,t}$ satisfies $\Gamma_t = \Gamma_s \Delta_{s,t}$. A stochastic divisor contracts total variation, so under P-divisibility $\|\delta_a \Gamma_t - \delta_b \Gamma_t\|_{\mathrm{TV}}$, with δ_a the point mass at a, cannot increase. This is the divisibility criterion of [RHP14, Thm 2.4] and the data-processing direction of [BD16, Prop. 1].

In words, the family is P-divisible when every later table of probabilities from the start can be obtained from every earlier one by one further table of transition probabilities; when it fails, for some earlier and later time no stochastic map carries the earlier rooted transition matrix to the later one, and when the earlier matrix is invertible the unique candidate has a negative entry (§2.6). Each row of Γ_t is the forecast from one possible starting state, its *root*. The hidden part is prepared the same way whatever is visible at the start, and independently of it: this is the hidden prior shared by every root. It matters because the table Γ_t is then a property of one physical setup, the same hidden preparation run forward from each visible start: the row for a start that occurs is the forecast the process gives from it, and a start that never happens still has a row, computed from the same preparation.

The step from two observed histories with a common endpoint to an *overlap*, two rows that are both positive in a common column (Theorem 3), uses that every row comes from the same preparation [Main §2.3]. Rows for starts that never happen are kept because a result about a rule has to cover every start the rule can be given, not only the starts a particular sample contains. The argument compares the whole square table with its return to certainty at the recurrence, and it can happen that the only rows keeping the table from being a permutation belong to starts that never happen, as in the different-deck example [Main §2.3]. The price is that the failure of chaining can then lie only in forecasts that no observer will see; when every start has positive probability, every row is one that occurs (see the Corollary).

Theorem 3 (recurrence and overlap [Main §2.3]). *Let φ be a bijection of $\mathcal{C}_V \times \mathcal{C}_H$ with one hidden prior shared by every root and drawn independently of it. If two rows of Γ_s have a positive entry in a common column and $\Gamma_n = I$ for some $n > s$, the family is not P-divisible on $\{0, \dots, n\}$.*

A step-by-step forecast can never move two forecasts further apart, so two forecasts that overlap at time s can never become the two distinct certainties of $\Gamma_n = I$.

Finite recurrence supplies the return, $\Gamma_L = I$ for the order L of φ, the recurrence time. The step from an exact microscopic iterate return to the actual rooted identity map, using the common normalized hidden prior, and its composition with the overlap theorem are machine-checked; existence of a finite return and the history-to-overlap implications in the following Corollary retain their separate written proofs. A *return horizon* is a time n with $\Gamma_n = I$, at which the forecasts from the start return to certainty, whether or not the whole world has returned; L is one, and there may be earlier ones. The horizon $\{0, \dots, L\}$ up to the world's return is its *recurrence cycle*:

Corollary (recurrence-cycle indivisibility). *Let φ have order L, with one hidden prior shared by every root and drawn independently of it, and let Γ_t keep a row for every root, roots of visible probability zero included. If some Γ_s with $0 < s < L$ is not a permutation, the rooted family is not P-divisible on $\{0, \dots, L\}$. Two distinct positive-probability visible histories of equal length*

with a common endpoint, in particular a memory witness, supply such a Γ_s, and so does every non-deterministic visible step, also when the law is Markov on every accessible window. With a full-support visible prior, this failure is equivalent to $M_k > 0$ at some order $k < L$.

So, in a finite reversible world whose hidden part starts with the same odds whatever is visible at the start, the whole system eventually returns to its starting configuration, and the forecasts from the start return to certainty with it. If at some earlier time the forecasts from two different starting points overlap, as happens at some time whenever a visible step is not fully predictable from the current observation, no chain of step-by-step forecasts can lead back to certainty, so the forecasts cannot be chained somewhere in the cycle.

The two halves of the Corollary fit together. A record is *memoryless on a window* when, at every step inside that window, the next observation depends only on the present one, so that $M_t = 0$ there. Conditional memory can exist all the same, at a later order, because the hidden part can hold a trace of the past that changes what is seen only later: the record then shows conditional dependence on the past at some order beyond the window but before the world returns to its start, and across the whole recurrence cycle that late dependence appears as the failure of chaining. Accessible windows are, by definition, short compared with the world's recurrence time L [Main §2.3, Scope], and the gap between them leaves room for conditional memory that no accessible window detects: a trace held in the hidden part can first affect what is seen at an order beyond every accessible window, though still before the return. Two cases differ. If the conditional memory first affects a step that some accessible window still reaches, an observer whose observations reach that step may detect it, as in the example below. If it first affects what is seen only beyond every accessible window, an observer confined to accessible windows sees neither the late conditional memory nor the failure of chaining it implies across the cycle: both remain established in the dynamics under the theorem's hypotheses, but neither appears in any accessible record. The failure of chaining is a property of the world's rule, its dynamics with its hidden preparation: it constrains how such a world can be described, not what accessible data will show [Main §2.3, Scope].

For example, let X_0, X_1, X_2 be independent fair bits and $X_3 = X_0$. Up to time 2 each next observation is independent of the past, so $M_1 = I(X_0; X_2 \mid X_1) = 0$: the record has no accessible memory on that window, and its forecasts from the start chain step by step there. Yet every deterministic world that reproduces the law, reversible or not, must keep X_0 in its hidden part at time 2, and the past predicts the observation at time 3 exactly, $M_2 = 1$ bit [Main §3.4, Remark (a variational question)]. Here the window ends at time 2: an observer whose observations reach time 3 may detect the memory, while one whose windows end at time 2 cannot detect it through this record. The example shows memory beyond the window; it shows no failure of chaining inside it.

Stated precisely: for the rooted family, with a hidden prior shared by every root and drawn independently of it and with a full-support visible prior, failure of P-divisibility on $\{0, \dots, L\}$ is equivalent to $M_k > 0$ at some order $k < L$ (the Corollary); a law that is Markov on every accessible window has $M_k = 0$ at every accessible order, so that k lies beyond the accessible windows. Picture the orders $1, 2, \dots, L-1$ as a long line: accessible windows cover only its short beginning, while the order k of the Corollary can lie anywhere on it, so a conditional memory that first appears late on the line is real in the dynamics but outside every window.

Recurrence-cycle indivisibility thus needs only finite reversibility and one non-deterministic visible step, and holds also for laws that are Markov on every accessible window, so it does not separate them from memory-bearing laws at accessible orders; what an observer can discriminate is memory at an accessible order, which (C4) asserts [Main §2.3, Scope].

Proofs and sources. *Theorem 3.* Two such rows are at total-variation distance below 1 at time s and exactly 1 at time n, which the contraction criterion forbids; Theorem 3 is machine-checked in its rooted form, and Doukas states the same argument concurrently [Dou26, Prop. 1]. Two rows of a square stochastic matrix share a positive column exactly when it is not a permutation matrix, and [Main §2.3] also proves the theorem from the lemma that a finite square stochastic matrix with a stochastic one-sided inverse is a permutation matrix [BB12, Problem 23] (machine-checked). Monotone distance is necessary for P-divisibility but not sufficient [Main §2.3].

The Corollary. This is [Main §2.3, Theorem (history readback + finite recurrence implies indivisibility) and Scope]: the first sentence is Theorem 3 with $n = L$, since two rows of a non-permutation Γ_s share a positive column, and Theorem 3 is machine-checked in its rooted form; the supply of a non-permutation Γ_s is proved on paper. For the equivalence, a memory witness below L gives a non-permutation Γ_s; conversely, with none the law on $\{0, \dots, L\}$ is Markov, $\Gamma_L = I$ makes every Γ_s a permutation by the stochastic-inverse lemma, and $\Gamma_s^{-1}\Gamma_t$ is a stochastic divisor, full support being needed only in this converse. Indivisibility at some return horizon is also machine-checked from the *routed* form of (C4), the realization clause in which a record of the visible history written into the hidden state is routed back into later visible conditionals within the window [Main §1.3].

2.6 Three examples

A minimal model. A hidden register sometimes flips the visible bit and remembers having done so, so that a second step undoes the first. Let $\mathcal{C}_V = \{0, 1\}$ and $\mathcal{C}_H = \{1, \dots, 6\}$ [Main §2.4]. For $h \in \{1, 2\}$, φ swaps $(0, h)$ with $(1, h)$: it flips the visible bit and keeps h, which then records the flip. At fixed x, φ swaps $h = 3$ with 4, and 5 with 6. A second application reads the record and flips the bit back. With h uniform,

$$\Gamma_1 = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{pmatrix}, \qquad \Gamma_2 = I, \qquad \Gamma_1^{-1}\Gamma_2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

The only candidate divisor has negative entries, so the family is P-indivisible, whereas a Markov chain with the same one-step matrix would give Γ_1^2, with diagonal entries $5/9$ and off-diagonal entries $4/9$. With x_0 uniform and independent of h, $P(x_2{=}0 \mid x_1{=}1, x_0{=}0) = 1$ while $P(x_2{=}0 \mid x_1{=}1) = 1/3$; for $P(x_0{=}0) = a$, the second value is $a/(2 - a)$.

Memory without short-window indivisibility. The last bit is the parity of the first two: neither earlier bit alone predicts it, and both together determine it. Let X_0 and X_1 be independent uniform bits and $X_2 = X_0 \oplus X_1$. Every two-time transition matrix is $\frac{1}{2}J$, with J the all-ones matrix, so the family factors through $\Delta = I$ and is P-divisible on $\{0, 1, 2\}$. Yet $I(X_0; X_2 \mid X_1) = 1$ bit. Up to flipping the final bit, this is the process of [RHP14, Eq. (15)]; it appears concurrently in [Dou26, App. A]. So on this window the forecasts from the start chain step by step, yet the record has conditional memory, and every reversible world that reproduces it carries that memory in its hidden part: memory in the record and P-divisibility are different properties. A reversible world that reproduces this law, with the hidden odds independent of the visible start, still fails to chain somewhere in its full cycle (§2.5), because its visible steps are not fully predictable; that failure need not lie on an accessible window.

The card table. The card analogy of §1.1 is also an exact finite model of the kind §§2.4–2.5 describe; as a picture of quantum mechanics it remains an analogy, since its statistics are classical (below). The two-card game of [Book §1.11] uses two cards, each red (0) or black (1): a visible *hole card* x, and a face-down *burn card* b, uniform and independent of x. One shuffle step swaps the two

cards, $\varphi(x, b) = (b, x)$. A *peek* reveals the hole card and overwrites the burn card, which holds the record, with a fresh card drawn uniformly and independently. Exact enumeration of the equally likely cases gives the probability that the hole card at the final reveal is the hole card as dealt:

Protocol	Probability
one shuffle	1/2
two shuffles	1
two shuffles, with a peek between them	1/2
Markov prediction for two shuffles, Γ_1^2	1/2

Here $\Gamma_1 = \frac{1}{2}J$ and $\Gamma_2 = I$. The two rows of Γ_1 are equal, so for every Δ the two rows of $\Gamma_1\Delta$ are equal and $\Gamma_1\Delta \neq I$: the family is P-indivisible.

The table separates the deterministic information from what the player can access. The shuffle is a bijection and loses nothing; the player sees only the hole card, and the memory the player observes, certainty after two shuffles where a Markov model predicts 1/2, is carried by the face-down card. The peek row depends on the rule: if the peek only reveals or replaces the hole card, the probability is 1, so the 1/2 comes from overwriting the record, not from looking. With anticorrelated cards ($b = 1 - x$), Γ_1 is a permutation and the game is divisible.

On card inputs the game is a classical stochastic process, and no card arrangement violates a Bell inequality [Book §1.11; Main §3.3].

3. Quantum emergence: scope and open steps

Section 2 shows that a quantum system measured in one fixed way and a partly hidden deterministic world produce the same patterns. Quantum mechanics says more. It allows measurements of different, mutually incompatible kinds, superpositions that are not mixtures of definite states, and systems combined into larger ones. The correspondence does not on its own give quantum mechanics, for three reasons: (1) it covers only records of one fixed measurement repeated at every step; (2) it holds for every law, so it selects neither the squared form of quantum probabilities nor the evolution relating two times (§3.3), and it represents the statistics without deriving a physical mechanism; (3) the further ingredients are five conditions that OI does not imply (§3.2).

3.1 Three notions kept apart

A record can have memory, can fail to chain, or can carry quantum memory, and the three can occur separately. A *quantum channel* is the most general way a quantum state can change, noise included; it is *completely positive* when it remains a valid change when applied to part of a larger system. *Tracing out* a system means ignoring it, a *diagonal* hidden state is one with definite configuration probabilities and no superposition, and the *populations* of a quantum state are the probabilities of its basis outcomes. The *Hadamard gate* is the basic operation that turns a definite 0 or 1 into an equal superposition of both.

The indivisibility of §2.5 is global: it occurs somewhere in the recurrence cycle, not on every window, and conditional memory does not force short-window indivisibility (§2.6). A revival of distinguishability within a window, two roots whose rooted laws are closer at an earlier time than at a later one, does make the family P-indivisible on that window [Main §2.3].

Three notions are distinct, and no unconditional implication among them is asserted [Main §3.2, Remark (three notions, kept apart)]: (1) accessible memory, $M_t > 0$; (2) stochastic P-indivisibility of the visible population law; (3) quantum non-Markovianity under a chosen criterion. Among what is proved: accessible memory does not force P-indivisibility on a window (§2.6), and outside the diagonal-preserving class P-indivisibility does not force CP-indivisibility (below). A family of quantum channels Φ_t is completely-positive (CP) divisible if $\Phi_t = \Psi_{t,s} \circ \Phi_s$ with each $\Psi_{t,s}$ completely positive and trace-preserving. For the family obtained by tracing out a permutation unitary from a diagonal hidden state, (2) implies CP-indivisibility [Main §3.2, Theorem (CP-indivisibility, framework scope)]. Outside the diagonal-preserving class the implication fails: with Φ_{t_1} conjugation by the Hadamard gate and $\Phi_{t_2} = \mathrm{id}$ the family is CP-divisible, while its computational-basis populations, $\frac{1}{2}J \to I$, are P-indivisible [Main §3.2, Remark (scope: P-indivisibility is not quantum non-Markovianity)].

So memory in the record, failure of step-by-step chaining and the quantum notion of memory are three different properties; the Hadamard example is a perfectly chainable quantum evolution whose recorded populations cannot be chained.

In the card table of §2.6, embedded as a permutation unitary with the burn card in the uniform diagonal state, the one-step channel $\rho \mapsto \mathrm{tr}(\rho)\, I/2$ is entanglement-breaking (measure-and-prepare) and the two-step map is the identity, so the family is CP-indivisible [Main §3.2]; its memory is still classical in the sense of [BBS24], since the two-step identity is obtained by realizing the one-step channel as the Pauli twirl $\rho \mapsto \frac{1}{4}\sigma_o \rho \sigma_o$, $o \in \{0, x, y, z\}$, and applying σ_o conditioned on o, a decomposition a fixed initial hidden card does not admit [Main §4.5].

That is, the embedded card game fails quantum chaining as well, yet its memory is ordinary classical memory, so failure of quantum chaining does not by itself make memory quantum in that sense.

Distinguishability backflow witnesses divisibility failure in one direction only [BLP09]; for bijective dynamical maps on a d-dimensional system, CP-divisibility is equivalent to monotone distinguishability with a $(d+1)$-dimensional ancilla [BJA17]. Theorem 1 states which laws admit realizations and quantum representations, not why a physical system exhibits a law with memory [Main §3.4, Remark (representation, not explanation)]; Calvo states a derivation of the textbook axioms from indivisible stochastic evolution [Cal26], whereas Theorem 1 derives no axioms.

3.2 From the correspondence to operational quantum mechanics

When are the operations an experimenter can perform on a finite system, with helper systems attached, exactly those of quantum mechanics, every kind of measurement included? For a theory with the structure described below, Theorem 4 answers with five requirements on what the experimenter can do. An *ancilla* below is an auxiliary system brought in for an operation, a *Kraus instrument* is the general quantum operation with outcomes, *coherence* is the capacity of a quantum state to be a superposition rather than a mixture of definite states, and a *qubit* is the simplest quantum system, with two basic states.

Fix a nonempty finite system and an operational theory specifying which finite outcome families of linear maps are available on the system and at every ancilla level, that is, with each finite ancilla adjoined. The theory contains the identity and is closed under coarse-graining of outcomes and, at every ancilla level, under outcome-conditioned composition. It has an available preparation of a uniformly mixed ancilla, which an available operation may follow, an available spectator-independent readout of the ancilla, one that acts on every product input as the identity on the

system, and ancilla discard: an available preparation followed by an available instrument, with the ancilla then discarded, is an available instrument on the system with the same outcomes [GR §3.3]. That is, the experimenter can always leave the system alone, combine operations, lump outcomes together, prepare an evenly mixed helper and act after preparing it, read the helper without disturbing the system, and discard it. The five completion conditions are:

(i) *valid probabilities*: available families send positive states to positive states, and their outcome traces sum to the input trace;
(ii) *trivial-ancilla consistency*: operations remain available with a one-state ancilla adjoined;
(iii) *inert spectators*: an untouched independent system can be adjoined without changing an intervention;
(iv) *full reversible control*: every finite unitary on every composite is available;
(v) *iterated composition*: attaching a fresh uniformly prepared ancilla to a composite, acting, and discarding it gives an available operation, so that a composite can itself serve as the working system of a larger experiment.

Theorem 4 (operational-completion characterization [GR §3.3]). *For every nonempty finite system, the available outcome families on the system and at every positive ancilla level are exactly the normalized finite Kraus instruments if and only if conditions (i)–(v) hold.*

That is, for a theory with the structure above, the operations available on a nonempty finite system, with any finite helper attached, are exactly the quantum ones (the normalized finite Kraus instruments) if and only if all five requirements hold: probabilities make sense; adding a trivial helper or an untouched bystander changes nothing; every reversible quantum operation on any combined system is available; and a fresh, evenly mixed helper can be added to any combined system, used and discarded.

Each condition is independent of the other four and of the observation process itself: for one fixed finite embedded-observation process satisfying (C1)–(C4), and for each condition, there is a qubit theory that realizes that process and the other four conditions, and fails that condition. In particular, a qubit theory that never creates coherence realizes the observation process and is not quantum, so OI alone does not imply quantum mechanics (both machine-checked) [GR §3.3]. Conditions (i) and (ii) only make the theory well formed; (iii)–(v) are the substantive requirements, which [GR §3.3] calls selection principles. Which of (iii)–(v) a physical substratum supplies is the open step toward full operational equivalence.

Proofs and sources. Both directions are machine-checked (Appendix A). The sufficiency direction rests on the Stinespring dilation and on the realization of every finite instrument by a unitary interaction with an ancilla followed by ancilla readout and discard [Sti55; Kra83; Oza84]; for reconstructions of finite quantum theory from operational axioms see [Har01; CDP11; MM11].

3.3 The evolution relating two times is not selected

Two different quantum mechanisms can produce identical records at every time, so the records alone cannot tell which evolution connects two moments. A *coherent lift* of a visible family is a quantum evolution of the visible and hidden parts together, a unitary *dilation*, that reproduces the visible law at every time; a *right action* changes a lift by a transformation applied before it.

Representability does not fix the unitary evolution relating two times [Main §3.4, Remark (scope: representability, not selection of the relating evolution)]. Two coherent lifts of one visible family, unitary dilations that reproduce the visible law at every time and compose exactly across times, can

differ by a time-dependent right action invisible to the visible law and then have different relating evolutions (machine-checked on an exhibited pair of lifts), while a constant deformation cancels from every relating evolution (machine-checked in general).

For lifts with trivial ancilla, read in a fixed basis, two lifts have equal rooted families exactly when their one-step visible matrices are equal, and equal one-step matrices give equal trajectory laws at every horizon; each such trajectory law is a fixed-basis Born record and stochastic (all machine-checked; the first is an equivalence). The fixed-basis correspondence therefore acts on the quotient by visible equality, and selecting the relating evolution needs a condition beyond representability and ordinary coherence.

So the fixed-basis records cannot distinguish two quantum descriptions with trivial ancilla that agree on everything visible, and choosing the evolution between two times needs a principle that the records do not supply.

3.4 No fixed finite realization reproduces a qubit

A finite hidden world has finitely many states, while a single qubit can be prepared and measured in infinitely many ways. Here the *ontic configurations* are the hidden world's states, a *prepare-and-measure* experiment prepares a system in one of several ways and then measures it in one of several ways, and its *behaviour matrix* lists the outcome probabilities for every preparation and measurement.

A fixed finite realization with N ontic configurations generates only behaviour matrices that factor nonnegatively through N states. Simulating a qubit's prepare-and-measure statistics needs classical memory unbounded over finite experiments: explicit tasks are given by Galvão and Hardy [GH03] and Montina [Mon08], and Montina reports Hardy's result that the number of ontic states cannot be finite [Mon08; Har04]. So no fixed finite realization reproduces the full prepare-and-measure theory of a qubit exactly; the factorization is proved, and the unboundedness is imported [Main §3.4, Proposition (finite-substratum operational obstruction — classical-dimension form); HKLP24]. The obstruction does not touch Theorem 1, whose realization is chosen per law.

That is: for any single finite pattern some finite hidden world reproduces it, but no one fixed finite hidden world reproduces everything one qubit can do across all possible experiments, which would need unlimited hidden memory.

3.5 Bell structure

Two laboratories far apart share a prepared pair of systems. Each chooses a setting, that is which measurement to make, and records an outcome $+1$ or -1. Bell's theorem asks whether the correlations between the two records can be explained by information the pair carried from the start, each outcome depending only on its own laboratory's setting. For any such explanation the Clauser–Horne–Shimony–Holt combination S_{CHSH} of the four setting-pair correlations is at most 2 in magnitude, while quantum mechanics reaches $2\sqrt{2}$. Each laboratory with its part of the pair is a *wing*, and a *pre-setting variable* is the hidden state before the settings are chosen. For a deterministic hidden world, Bell's theorem tests directly what partial access can and cannot explain.

[Main §3.3] proves a Bell ceiling under the framework's own hypotheses. The spatial substratum is a product of finite site factors, and its coupling graph joins site i to site j when the ith output component of φ depends on the jth input component, and d_G is its graph distance; an intervention on a region A changes, k updates later, only components within d_G-distance k of A [Main §3.3].

Let Ω be a pre-setting variable that fixes each wing's response to every setting, and suppose that (i) the dynamics is deterministic; (ii) the setting operation acts at one step on disjoint regions A and B, each intervention reading and writing only its own region; (iii) wing A's outcome is a ± 1-valued function of the state on a readout region R_A, k_A updates after the setting, with $d_G(B, R_A) > k_A$, and symmetrically $d_G(A, R_B) > k_B$; and (iv) the interventions act on one common pre-setting ensemble, $p(\Omega \mid a, b) = p(\Omega)$. If the protocol records a ± 1 outcome in every run, the model is Bell-local, with deterministic local response functions, and the Clauser–Horne–Shimony–Holt combination obeys $|S_{\mathrm{CHSH}}| \leq 2$ [Bel64; CHSH69; Main §3.3, Theorem (Bell ceiling for a screened local completion)]. Reproducing the quantum values $2 < |S_{\mathrm{CHSH}}| \leq 2\sqrt{2}$ therefore requires violating at least one of these assumptions.

That is, a deterministic world whose two laboratories cannot affect each other's results within the time of the experiment, whose settings are chosen independently of its hidden state, and which records an outcome in every run, cannot produce quantum Bell correlations.

In the framework, determinism is part of OI and the protocol records an outcome in every run, so a completion reproducing quantum Bell correlations gives up at least one of the remaining assumptions: (ii) and (iii), which with the causal-cone property secure ontic parameter independence, or (iv), measurement independence [Main §3.3]. *Ontic parameter independence* means that in the underlying (ontic) state one wing's response does not depend on the other wing's setting; *measurement independence* means that the settings are statistically independent of the pre-setting state. The framework retains measurement independence, as a posit about the initial condition, and adopts ontic parameter dependence: one wing's microstate response may depend on the remote setting, with operational no-signalling required. The nearest-neighbour cubic update of the lattice substratum [SM] alone cannot supply that dependence for wings outside each other's causal cone [Main §3.3, Corollary (Bell–lattice obstruction and state-dependent escape)]. That a preparation-indexed coupling graph, in which a Bell preparation adds bounded-degree long-range edges to the coupling graph, can supply it while preserving operational no-signalling and the gravitational continuum limit is the open hypothesis H-Bell [Main §3.3].

So, given determinism and an outcome in every run, a deterministic hidden world can produce quantum Bell correlations only if one laboratory's hidden response depends on the other laboratory's choice, or if the choices are correlated in advance with the hidden state. The framework keeps the choices independent and accepts the first, with no usable signal allowed between the laboratories; whether a coupling graph on the lattice substratum can supply that dependence without any usable signal and while keeping the gravitational continuum limit is open (H-Bell). Here the coupling graph records which parts of the substratum directly affect which, and H-Bell is the named hypothesis that suitable extra links can be added for a Bell preparation with those two properties.

4. The cosmological constant: how OI changes the interpretation of vacuum energy

Quantum theory gives empty space an energy: every field keeps a minimum jitter even in its lowest state, and adding up the energy of that jitter gives an enormous number. Gravity responds to energy: in general relativity the *stress-energy*, the energy and momentum content of matter and fields, sources the curvature of spacetime, so if the vacuum energy counted in full, space would curve far more strongly than is observed. The question matters because the vacuum-energy estimate exceeds the observed value by some 120 orders of magnitude [Wei89, §III], and OI changes how the two quantities are related. In OI the quantum description is *emergent*: it is the description of

the observer's visible statistics, not of the underlying world. A *Hubble-scale density* is the energy density set by the present expansion rate of the universe, and the *de Sitter entropy* is the entropy assigned to the horizon of a universe expanding at that rate.

Summing zero-point energies of the quantum fields up to a Planck-scale cutoff, the smallest length at which present physics is expected to apply, gives a vacuum energy density some 120 orders of magnitude above the observational value, the exact power depending on the cutoff convention [Wei89, §I; §III, Eqs. (3.4)–(3.6)]. If that density sources curvature, it must be cancelled to that accuracy [Wei89, §III]; this discrepancy is here called the mismatch. With a Planck-scale cutoff and a Hubble-scale density the mismatch equals the de Sitter entropy up to a numerical factor fixed by the cutoff convention, a dimensional identity [GR §6.3].

What OI establishes. In OI the zero-point sum belongs to the emergent quantum description (§§2–3), while the effective cosmological constant Λ belongs to the gravitational description of the substratum, with its classical stress-energy $T^{\mathrm{cl}}_{\mu\nu}$: the two are defined at different levels. A uniform shift $\hat{H} \to \hat{H} + C$ of the emergent Hamiltonian changes no visible transition probability at any precision, so the zero-point offset is absent from every visible statistic of the quantum description [GR §6.2]. That no non-gravitational measurement sees the absolute vacuum energy is familiar; what OI adds is the second level, at which a gravitational source can be defined independently of the emergent vacuum.

So in OI the vacuum jitter belongs to the observer's quantum description, and the cosmological constant to the description of the underlying world; shifting every energy of the quantum description by the same constant changes no observable probability. That gravity responds only to the underlying world is the proposal below, and deciding it requires the field equations.

The proposed resolution. If the covariant field equations take the form [GR] targets,

$$\mathcal{G}_{\mu\nu} = 8\pi G\,(T^{\mathrm{cl}}_{\mu\nu} - \mathcal{F}[g, T^{\mathrm{cl}}]\,g_{\mu\nu}),$$

with $\mathcal{G}_{\mu\nu}$ the gravitational side, still to be derived (the Einstein tensor in general relativity), and $\mathcal{F}$ a scalar functional of the metric and the stress-energy, then the zero-point sum of the emergent description is not a source of curvature and the mismatch requires no cancellation. The vacuum term that remains, from the substratum's classical energy density and from $\mathcal{F}$, is not computed, and the observed value of Λ enters as a cosmological input.

The missing link. The quantum description alone cannot decide whether the offset sources curvature. The completions

$$\mathcal{G}_{\mu\nu} = 8\pi G\,T^{\mathrm{cl}}_{\mu\nu} \qquad \text{and} \qquad \mathcal{G}_{\mu\nu} = 8\pi G\,(T^{\mathrm{cl}}_{\mu\nu} - \rho_C\,g_{\mu\nu}),$$

with ρ_C the energy density of the offset C, share the substratum, the partition and every visible transition law [GR §6.2]. Deciding between them requires covariant field equations derived from the substratum, with their conservation and coupling to sources, which [GR] calls the G3 coupling theorem and leaves open [GR §6.2, §8.5]; the same step fixes how the energy of electroweak and QCD vacuum transitions gravitates. A derivation of these equations would turn the proposal into a theorem or refute it.

Trace-free Einstein equations [EvEMU11] and vacuum-energy sequestering [KP14] also remove matter vacuum energy as a source of curvature, and Volovik argues that the ultraviolet zero-point sum does not set the vacuum energy [Vol06]; OI locates the gravitational source in the substratum microstate.

5. Black-hole information: preservation versus accessible recovery

The black-hole information problem asks whether the radiation of an evaporating black hole carries the information of what formed it [Haw76; Pag93; Pag13]. In OI two questions separate: whether information is conserved by the global dynamics, which holds because the dynamics is a bijection, and whether it can be recovered from the records an observer can access [HP07]. Finite exact computations show that these are different properties. As an analogy, a burned letter is not destroyed in principle: the smoke, ash and heat still encode its words, because the underlying laws lose nothing, yet nobody who sees only the smoke can read it. The question matters because the problem is usually posed as one of conservation, and OI shows that conservation and recovery must be answered separately.

The toy model [BHa]. The configuration space is a finite product of one-bit registers. A message m, drawn uniformly from a set $\mathcal{M}$, is written into designated interior registers; every other register is uniformly random and independent of m, or blank. One map is applied at every step, and at each *emission* one register passes from the interior to the radiation. The observer reads registers on a schedule fixed in advance and keeps every read; for the first count below, it reads the radiation registers as they are emitted. The measured quantity is P_{succ}, the probability that the best guess of m from these records is correct, and the *accessible advantage* is its excess over the blind value $1/|\mathcal{M}|$. $\mathcal{U}$ is the set of joint values of the registers left unread at the end. The layout is that of the Page-curve setup [GR App. A.7] at small size.

Results [BHa]. Conservation does not imply recovery: of the 40,320 bijections of $\{0,1\}^3$ with one message bit and two random bits, 1,376 give zero accessible advantage with one register unread after two emissions. Nor is it necessary: 60 non-injective maps of $\{0,1\}^2$ recover the message exactly when every register is read. What bijectivity guarantees is a floor: reading every register at the end gives exact recovery, and with $\mathcal{U}$ unread $P_{\text{succ}} \geq 1/|\mathcal{U}|$, so zero advantage requires $|\mathcal{U}| \geq |\mathcal{M}|$; 32 non-injective maps of $\{0,1\}^2$, with a two-bit message occupying both registers and one register read, violate this floor. It is the remnant-capacity form of the fact that $|\mathcal{U}|$-valued side information multiplies a guessing probability by at most $|\mathcal{U}|$, so hiding the message completely needs at least as many hidden values as messages, as in perfect secret sharing, of which Shamir's threshold scheme is the standard example [Sha79]. The card table of §2.6 shows the same separation in its simplest form.

So a reversible world keeps every message, but keeping is not showing. In some reversible worlds (1,376 of the 40,320 toy maps) a message stays completely hidden from an observer who reads only the radiation, which is possible only when the unread registers have at least as many joint values as there are messages; and a world that loses information can still show a message exactly.

The *entanglement entropy* of the radiation measures how strongly it is correlated with what remains of the black hole; the *Page curve* is its expected course, rising and then falling back to zero as the black hole finishes evaporating, if the information comes out. An *energy shell* is the set of configurations with a given energy; the *orbit ensemble* is the set of states the dynamics passes through; the *bipartition* is the split of the system into black hole and radiation; and *measure concentration* is the fact that in very large state spaces almost every state gives nearly the same value of a smooth quantity such as the entanglement entropy.

The Page-curve obstruction. The calculation of [GR App. A] uses $\mathcal{C}_V \times \mathcal{C}_B \times \mathcal{C}_D$, with V the exterior, B the black-hole interior, D the sector beyond the cosmological horizon, and radiation $R \subset V$ emitted from B. Its three conditions are H-scramble (the orbit ensemble explores the

required quantum pure states), H-decouple (a pure $B \cup R$ evolves uncoupled from D and $V \setminus R$ over whole cycles), and constant energy on all of $\mathcal{C}_B \times \mathcal{C}_R$, not merely energy conserved along an orbit. In the theorem's stated configuration-state ensemble and coordinate-aligned bipartition, each permutation maps a product basis state to another product basis state. H-scramble therefore has no instance, and Theorems A.8–A.9 as stated do not establish a nonvacuous Page curve [GR App. A.7–A.9; BHb].

There are independent limits on the proposed black-hole application. The effective-pure-state result [GR, Thm A.7] assumes that D initially has a definite configuration and controls $V \cup B$ only for times short compared with the D-sector timescale $\tau_B^D \sim 1/H$; it does not establish that $V \setminus R$ is unentangled with $B \cup R$ over an evaporation cycle. An evaporating black hole also occupies a proper energy shell rather than the whole configuration product at one energy. A meaningful recovery needs an explicitly specified coherent preparation or a different partition, a nonvacuous scrambling condition and an entropy/ensemble statement appropriate to that model. That is an open research problem, distinct from the finite conservation-versus-recovery results above.

A nonvacuous finite entropy curve. Coherent preparations admit a restricted mathematical construction [CE]. Prepare k binary registers in $|+\rangle$ and $n-k$ in $|0\rangle$, choose one uniformly random invertible binary linear map, and apply its reversible configuration permutation. The resulting pure state is the uniform coherent superposition on a uniformly chosen k-dimensional subspace $W \subseteq \mathbb{F}_2^n$. At the nested coordinate cut $R_r = \{1, \ldots, r\}$, with complement B_r, its entanglement entropy in bits is exactly $S_r = k - \dim(W \cap R_r) - \dim(W \cap B_r)$. For $n = 2m$ and $k = m$, the same sampled state at every cut has zero entropy at both endpoints and mean entropy between $\max(0, \min(r, n-r) - 2)$ and $\min(r, n-r)$. The finite subspace-counting proof and independent exact controls establish this ensemble statement; it is not Page's Haar mean.

A companion construction [CR] turns this cut profile into an actual reversible toy trajectory: a clock in a definite configuration controls successive swaps from an interior register to initially blank radiation registers. During the first release window the radiation's entropy is exactly the corresponding S_r, for the same prepared W throughout. Mirror Bell-pair preparation gives the exact tent $\min(r, n-r)$; a configuration input or an unmixed coherent product gives zero throughout. The reversible clock later transfers the state back. These are written finite proofs, not Lean certificates or a black-hole evaporation model: the coherent preparation and ensemble are stipulated, and energy-shell dynamics, gravitational coupling, physical scrambling and permanent radiation decoupling remain open. They supply a nonempty finite recovery example while leaving the configuration-state obstruction intact.

6. Established results, conditional interpretations and open problems

Summary. OI asks whether quantum randomness could come from an observer's partial view of a deterministic world, and the mathematics answers part of that question exactly. Over any finite number of steps, every pattern of fixed-measurement quantum records is what some partly hidden deterministic world shows. When the record has memory, the memory is held in the hidden part, which must be large enough to hold it; conversely, if the hidden part of any one such world holds a trace of the past that changes what is seen next, the record has memory. In a finite reversible world whose hidden part starts with the same odds whatever is visible at the start, any visible step that is not fully predictable forces, somewhere in the world's cycle, forecasts from the starting points that cannot be chained step by step. For an operational theory with the structure of §3.2 on a nonempty finite system, the available operations, on the system and with every finite ancilla attached, are

exactly the normalized finite Kraus instruments, incompatible measurements included, if and only if five stated conditions hold, and a physical substratum is not shown to meet them; nor is the squared form of quantum probabilities selected. Each of these is a statement about what partial access to a deterministic world does and does not produce. In the applications OI separates two levels of description for vacuum energy and separates keeping information from showing it for black holes; the gravitational field equations remain open, and the configuration-state Page theorem has no instances. Hidden predictive memory does not imply permanent hidden-state uncertainty: identification is relative to specified records and operations [Substratum, Prop. 24.5].

Established mathematical results. Machine-checked: Theorem 1; the cores of Theorem 2 and of the capacity bound, their actual finite-trajectory information bounds and the supremum over all finite future blocks; Theorem 3 in its rooted form, the actual-iterate-return bridge and the nonempty record-fibre identification core; Theorem 4 and the independence of its conditions; the non-selection of the relating evolution. Proved on paper, with Theorem 3 as machine-checked core and the shared-column fact and existence of finite recurrence as elementary steps: recurrence-cycle P-indivisibility of the rooted visible family, with a row for every root including roots of visible probability zero and a hidden prior shared by every root and drawn independently of it, whenever some rooted transition matrix is not a permutation; that every non-deterministic visible step and every memory witness supply such a matrix, and that with a full-support visible prior the indivisibility is equivalent to $M_k > 0$ at some order k below the recurrence time (§2.5). Proved on paper also: the remaining conditional-pushforward and total-variation time-t arguments of Theorem 2, its Corollary and the least hidden predictive memory; the Bell ceiling; the Bayesian support and passive stabilization parts of [Substratum, Prop. 24.5]; and the finite coherent-subspace entropy and sequential-release constructions [CE; CR]. The qubit obstruction combines a proved factorization with an imported unboundedness, and the separation of conservation from recovery rests on exhaustive enumeration with the $1/|\mathcal{U}|$ floor proved. Together these results establish that stochastic, quantum-like observations can arise exactly from deterministic reversible dynamics, and they locate what is not universal: the memory a law carries, its size and position in the hidden sector, and its consequence for divisibility. The correspondence represents these observations; it does not derive their physical mechanism.

Conditional physical interpretations. If OI holds as physics (with the bijective substratum an empirically motivated representation choice, §1.1), the fixed-basis statistics of quantum theory are the visible laws of an embedded observer, and their departures from Markov behaviour require hidden predictive memory, which may nevertheless be identified by retained records, as the card table of §2.6 illustrates. Given determinism and an outcome in every run, quantum Bell correlations require giving up ontic parameter independence or measurement independence; with measurement independence retained, they require ontic parameter dependence, whose realization on the substratum without usable signalling and with the gravitational continuum limit is the open hypothesis H-Bell (§3.5); the zero-point sum is not a gravitational source if the field equations take the targeted form (§4); and the finite coherent entropy-curve examples do not establish a physical Page curve beyond the configuration-state obstruction (§5).

Open problems. For quantum emergence:

1. full operational equivalence: which of completion conditions (iii)–(v) a physical substratum supplies (§3.2);
2. a selection principle for the quadratic Born exponent (§2.3) and for the evolution relating two times (§3.3);
3. history readback (C4) at the cosmological partition [GR §2.2], and an experiment or memory-

witness test distinguishing OI from standard quantum theory [Main §4.5];

4. the dynamical construction problems: exact gluing of per-horizon realizations for recurrent target families, approximation of a target on windows short compared with the recurrence time by one fixed substratum [Main §3.4], and the hidden-sector mixing hypothesis, motivated by eigenstate thermalization, under which observed conditional memory gives a lower bound on the hidden sector's coarse-grained mixing time, the slow-bath form of (C2) [Main §1.3, §3.4];
5. machine checks of the steps of §§2.2–2.5 proved on paper.

For the applications: a derivation of the covariant field equations with their source coupling, on which both §4 and, in [GR], the Page curve depend; a physical coherent preparation and scrambling/entropy theorem beyond the finite examples, and H-decouple; the proper-shell case, which an evaporating black hole whose radiation carries away its energy occupies; the entropy along a single trajectory [BHb]; the generalized second law [GR App. A.6]; and the relation to quantum-extremal-surface and island computations [Pen20; AEMM19; AMMZ20; Book §7.6] (§5).

Code, data and formal verification

The baseline declarations in Appendix A were checked in Lean 4.33.0 with Mathlib v4.33.0 at commit `17b830bc` by building `verification/lean-mathlib` (`lake exe cache get`, `lake --rehash build`) and running `python3 tools/release_gate.py`, whose `lean-axioms` step requires every `#print axioms` report to lie within `propext`, `Classical.choice` and `Quot.sound`; each declaration reports exactly these three, and the expected result and comparison recipe are in `papers/introduction_repro/`. The full commit hash is `17b830bcd0ecc36e10eff5b9eeca03bdd7198d86`. This pin applies to the baseline declarations, not the additional record-fibre, actual-return and trajectory-memory modules. Those modules are `OIBridge.RecordFibreIdentification`, `OIBridge.RootedRecurrenceBridge` and `OIBridge.TrajectoryMemory`, from source checkpoints `4f962b6`, `db658fd3` and `a958bc54`, respectively; their exact assumptions are summarized below. The corpus sources are distinguished in the References.

The probes and the black-hole computations [BHa; BHb] are released under the MIT License, at `papers/oi_lattice_code/foundations/` (probes; also in the release archive, concept DOI 10.5281/zenodo.19060318) and `verification/programmes/black-hole-information/` (not part of the release archive) at source snapshot `55e3217b`; the finite coherent-entropy and sequential-release controls [CE; CR] accompany their written proofs in the repository.

Appendix A. Machine-checked declarations

Modules of the `OIBridge` Lean library; baseline line numbers at commit `17b830bc`. The explicitly named additional modules are at the checkpoints given above.

- **Theorem 1** (§2.2): `finite_horizon_equivalence` (Equivalence:462); directions `S_imp_D` (:422), `D_imp_Qfb` (:294), `Qfb_imp_S` (:245), compositions :466–468; supporting `permMatrix_mem_unitaryGroup` (EquivalenceChain:179). Prior independence is proved on paper.
- **Theorem 2** (§2.4): `pushforward_identity`, `distinguishability_floor`, `capacity_floor`, `unavoidable_hidden_predictive_memory` (HiddenMemory:83, :114, :170, :185), for a joint law of history, present and hidden state; the conditional-pushforward/total-variation time-t arguments and the standard vanishing criterion for conditional mutual information [CT06] are proved on paper; the actual trajectory information bound is listed separately below.
- **Capacity bound and least hidden memory** (§2.4): `c3_necessity`, `c3_necessity_via_hidden`, `card_hidden_ge_two_pow_Istar` (C3Necessity:80, :88, :117), for bijective realizations in the same joint-law

form.

- **Stochastic-inverse lemma; Theorem 3** (§2.5): `isPermMatrix_of_stochastic_inverse` (EquivalenceChain:110); `pIndivisible_of_rooted_identity_return_of_overlap` (RecurrenceHorizon:148).
- **Recurrence-cycle indivisibility** (§2.5): `routed_forces_indivisible_somewhere` (PhysicalC4Discharge:122), indivisibility at some return horizon from routed readback; on paper for history-level (C4).
- **Actual trajectory memory and capacity** (§2.4): `next_joint_law`, `future_joint_law`, `future_at`, `trajectory_memory`, `trajectory_c3`, `trajectoryIstar_le`, `trajectory_capacity` (TrajectoryMemory). These push the initial law through actual permutation iterates and bound the supremum over all natural time/finite-horizon pairs; no literal infinite-tail identification is certified.
- **Nonempty record-fibre identification** (§§1.1, 2.4): `determined_iff_constant`, `initial_determined_iff_subsingleton`, `determined_injective_iff`, `current_determined_iff`, `current_visible_eq`, `current_hidden_determined_iff`, `swap_two_records_injective`, and `empty_record_counterexample` (RecordFibreIdentification). These do not prove the Bayesian support calculation or the $N-1$ stabilization bound.
- **Actual microscopic return** (§2.5): `rootedMap_eq_one_of_iterate_return`, `pIndivisible_of_iterate_return_of_overlap` (RootedRecurrenceBridge), conditional on an exact iterate return. These do not formalize finite-order existence, the common-endpoint history implication or an accessible-window witness.
- **Relating evolution** (§3.3): `gl2_strong_gauge_moves_relative_candidate` (CoherentLiftGauge:537, an exhibited pair of lifts), `gl3_constant_gauge_preserves_relative` (:451, general); `bridge_traj`, `rooted_eq_iff_slice_eq` (both directions: rooted families equal exactly when one-step matrices are), `realData_traj_stochastic` (TrackBQfbBridge:276, :180, :286).
- **Theorem 4** (§3.2): `exactAll_iff_physical_general` (GeneralCarrier:100); ⇒ `physical_of_exactAll` (PhysicalCharacterization:301); ⇐ `exactAll_of_physical_general` (GeneralCarrier:93) via `krausSoundExt_of_validity_inert` (OperationalValidity:124), `compositeCompleteness_unconditional` (IsometryExtension:179, premise discharged by `finiteIsometryExtensionSF_discharged`, :116) and `exactAll_of_levelOne` (LevelOneSeam:196).
- **Independence; OI alone does not imply quantum mechanics** (§3.2, qubit): `five_way_minimality` (RankGapTheory:702), `oi_alone_not_qm` (GeneralCarrier:160).

References

Corpus (concept DOI 10.5281/zenodo.19060318; Main, SM and the underlying black-hole computations at source snapshot `20ca05fe`; GR and the Book include the scoped SC-36 reconciliation from `021850d9` and the Book regress correction from `a2be1e45`)

[Main] A. Maybaum, "The Incompleteness of Observation: The Equivalence of Quantum Mechanics and Embedded Observation," `papers/Main.md`.

[GR] A. Maybaum, "$\hbar$, the Bekenstein-Hawking Entropy, and the Running-Vacuum Form of Dark Energy from the Cosmological Horizon," `papers/GR.md`.

[SM] A. Maybaum, "The Standard Model from a Cubic Lattice," `papers/SM.md`.

[Substratum] A. Maybaum, "The Substratum Construction: Reconstruction, the Substratum Gauge Group, and the QM-GR Synthesis," `papers/Substratum.md`; the record-fibre and restricted-invariance statements used here are at source snapshot `20ca05fe9e7e460de29e08f7a21029042fa45b12`.

[Book] A. Maybaum, *The Incompleteness of Observation*, `book/`.

[BHa] A. Maybaum, "Recovery from accessible records in finite reversible models," unpublished research record; proofs and code at `bh01_access_v1/` in the black-hole computations' directory.

[BHb] A. Maybaum, "Permutation dynamics versus quantum scrambling; Page-time fraction," unpublished research record; proofs and code at `bh02_scramble_v1/` in the black-hole computations' directory.

[CE] A. Maybaum, "Finite coherent-subspace entropy profiles," written finite proof and exact controls, `verification/audits/manuscript/page-coherent-recovery-20261028/entropy-profile-proof.md`, source checkpoint `bdcbd32`.

[CR] A. Maybaum, "Autonomous reversible sequential release," written finite proof and exact controls, the companion `sequential-release.md` in the same directory, source checkpoint `2624d49`.

External

[AEMM19] A. Almheiri, N. Engelhardt, D. Marolf, and H. Maxfield, "The entropy of bulk quantum fields and the entanglement wedge of an evaporating black hole," *J. High Energy Phys.* **12**, 063 (2019); arXiv:1905.08762.

[AMMZ20] A. Almheiri, R. Mahajan, J. Maldacena, and Y. Zhao, "The Page curve of Hawking radiation from

semiclassical geometry," *J. High Energy Phys.* **03**, 149 (2020); arXiv:1908.10996.

[Bar23] J. A. Barandes, "The Stochastic-Quantum Theorem," arXiv:2309.03085v2 (2023).

[Bar25] J. A. Barandes, "The Stochastic-Quantum Correspondence," *Philosophy of Physics* **3**, 8 (2025); arXiv:2302.10778v3.

[Bar26] J. A. Barandes, "The Born Representation Theorem and the Unistochastic Theorem," arXiv:2608.04354v1 (2026).

[BB12] J. C. Baez and J. Biamonte, "Quantum techniques for stochastic mechanics," arXiv:1209.3632v5 (2012).

[BBS24] C. Bäcker, K. Beyer, and W. T. Strunz, "Local disclosure of quantum memory in non-Markovian dynamics," *Phys. Rev. Lett.* **132**, 060402 (2024); arXiv:2310.01205v2.

[BD16] F. Buscemi and N. Datta, "Equivalence between divisibility and monotonic decrease of information in classical and quantum stochastic processes," *Phys. Rev. A* **93**, 012101 (2016); arXiv:1408.7062v3.

[Bel64] J. S. Bell, "On the Einstein Podolsky Rosen paradox," *Physics Physique Fizika* **1**, 195 (1964).

[BJA17] B. Bylicka, M. Johansson, and A. Acín, "Constructive method for detecting the information backflow of non-Markovian dynamics," *Phys. Rev. Lett.* **118**, 120501 (2017); arXiv:1603.04288v2.

[BLP09] H.-P. Breuer, E.-M. Laine, and J. Piilo, "Measure for the degree of non-Markovian behavior of quantum processes in open systems," *Phys. Rev. Lett.* **103**, 210401 (2009); arXiv:0908.0238v2.

[Cal26] S. Calvo, "On the stochastic-quantum correspondence," *Quantum Stud. Math. Found.* **14**, 2 (2026), doi:10.1007/s40509-026-00415-9; arXiv:2601.18720v2.

[CDP11] G. Chiribella, G. M. D'Ariano, and P. Perinotti, "Informational derivation of quantum theory," *Phys. Rev. A* **84**, 012311 (2011).

[CHSH69] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, "Proposed experiment to test local hidden-variable theories," *Phys. Rev. Lett.* **23**, 880 (1969).

[CT06] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. (Wiley, 2006).

[dMU21] L. de Moura and S. Ullrich, "The Lean 4 theorem prover and programming language," in *Proc. CADE 28* (Springer, 2021), p. 625.

[Dou26] J. Doukas, "On the emergence of quantum mechanics from stochastic processes," arXiv:2602.22095v3 (2026).

[EGGH26] G. Egri, M. Gömöri, B. Gyenis, and G. Hofer-Szabó, "Trajectory of probabilities, probability on trajectories, and the stochastic-quantum correspondence," arXiv:2602.23491v1 (2026).

[EvEMU11] G. F. R. Ellis, H. van Elst, J. Murugan, and J.-P. Uzan, "On the trace-free Einstein equations as a viable alternative to general relativity," *Class. Quantum Grav.* **28**, 225007 (2011); arXiv:1008.1196v2.

[GH03] E. F. Galvão and L. Hardy, "Substituting a qubit for an arbitrarily large number of classical bits," *Phys. Rev. Lett.* **90**, 087902 (2003); arXiv:quant-ph/0110166v3.

[Har01] L. Hardy, "Quantum theory from five reasonable axioms," arXiv:quant-ph/0101012 (2001).

[Har04] L. Hardy, "Quantum ontological excess baggage," *Stud. Hist. Philos. Mod. Phys.* **35**, 267 (2004).

[Haw76] S. W. Hawking, "Breakdown of predictability in gravitational collapse," *Phys. Rev. D* **14**, 2460 (1976).

[HKLP24] T. Heinosaari, O. Kerppo, L. Leppäjärvi, and M. Plávala, "Simple information processing tasks with unbounded quantum advantage," *Phys. Rev. A* **109**, 032627 (2024); arXiv:2308.07727v4.

[HLW06] P. Hayden, D. W. Leung, and A. Winter, "Aspects of generic entanglement," *Commun. Math. Phys.* **265**, 95 (2006); arXiv:quant-ph/0407049.

[HP07] P. Hayden and J. Preskill, "Black holes as mirrors: quantum information in random subsystems," *J. High Energy Phys.* **09**, 120 (2007); arXiv:0708.4025.

[KP14] N. Kaloper and A. Padilla, "Sequestering the Standard Model vacuum energy," *Phys. Rev. Lett.* **112**, 091304 (2014); arXiv:1309.6562v2.

[Kra83] K. Kraus, *States, Effects, and Operations*, Lecture Notes in Physics **190** (Springer, 1983).

[LP17] D. A. Levin and Y. Peres, with contributions by E. L. Wilmer, *Markov Chains and Mixing Times*, 2nd ed. (American Mathematical Society, 2017).

[Mathlib20] The mathlib Community, "The Lean mathematical library," in *Proc. ACM SIGPLAN Conf. Certified Programs and Proofs* (ACM, 2020), p. 367.

[Milz20] S. Milz, D. Egloff, P. Taranto, T. Theurer, M. B. Plenio, A. Smirne, and S. F. Huelga, "When is a non-Markovian quantum process classical?," *Phys. Rev. X* **10**, 041049 (2020); arXiv:1907.05807v3.

[MM11] Ll. Masanes and M. P. Müller, "A derivation of quantum theory from physical requirements," *New J. Phys.* **13**, 063001 (2011).

[Mon08] A. Montina, "Exponential complexity and ontological theories of quantum mechanics," *Phys. Rev. A* **77**, 022104 (2008); arXiv:0711.4770v2.

[Oza84] M. Ozawa, "Quantum measuring processes of continuous observables," *J. Math. Phys.* **25**, 79 (1984).

[Pag93] D. N. Page, "Average entropy of a subsystem," *Phys. Rev. Lett.* **71**, 1291 (1993); arXiv:gr-qc/9305007v2.

[Pag13] D. N. Page, "Time dependence of Hawking radiation entropy," *J. Cosmol. Astropart. Phys.* **09**, 028 (2013); arXiv:1301.4995v3.

[Pen20] G. Penington, "Entanglement wedge reconstruction and the information paradox," *J. High Energy Phys.* **09**, 002 (2020); arXiv:1905.08255v3.

[PSW06] S. Popescu, A. J. Short, and A. Winter, "Entanglement and the foundations of statistical mechanics," *Nature Phys.* **2**, 754 (2006).

[RHP14] Á. Rivas, S. F. Huelga, and M. B. Plenio, "Quantum non-Markovianity: characterization, quantification and detection," *Rep. Prog. Phys.* **77**, 094001 (2014); arXiv:1405.0303v2.

[Sch21] H.-J. Schmidt, "Dilation of stochastic matrices by coarse graining," arXiv:2106.03513v1 (2021).

[SGD19] P. Strasberg and M. García Díaz, "Classical quantum stochastic processes," *Phys. Rev. A* **100**, 022120 (2019); arXiv:1905.03018v2.

[Sha79] A. Shamir, "How to share a secret," *Commun. ACM* **22**, 612 (1979).

[Sti55] W. F. Stinespring, "Positive functions on C*-algebras," *Proc. Amer. Math. Soc.* **6**, 211 (1955).

[tH14] G. 't Hooft, "The cellular automaton interpretation of quantum mechanics," arXiv:1405.1548v3 (2014).

[Vol06] G. E. Volovik, "Vacuum energy: myths and reality," *Int. J. Mod. Phys. D* **15**, 1987 (2006); arXiv:gr-qc/0604062v4.

[Wei89] S. Weinberg, "The cosmological constant problem," *Rev. Mod. Phys.* **61**, 1 (1989).